GPIO Library Installation

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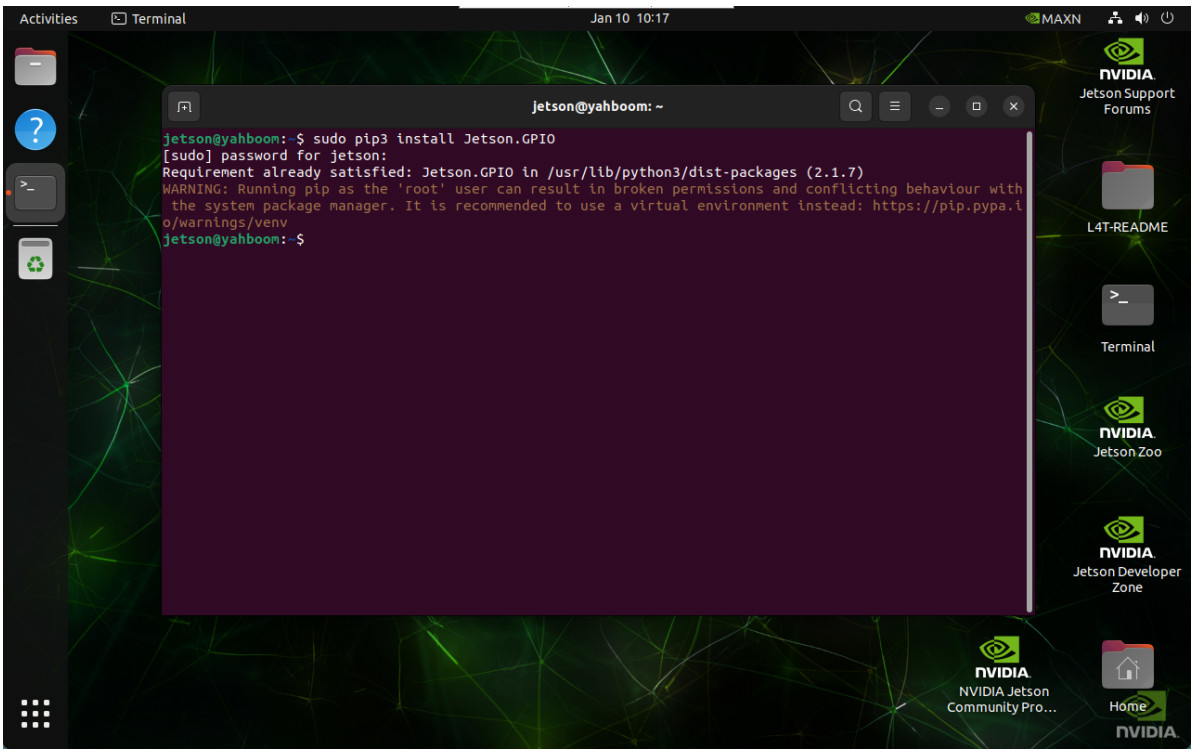
1. Installing the Jetson.GPIO Library

The system installs the Jetson.GPIO library by default; this step can be skipped.

1.1 Automatic Installation

```
sudo pip3 install Jetson.GPIO
```



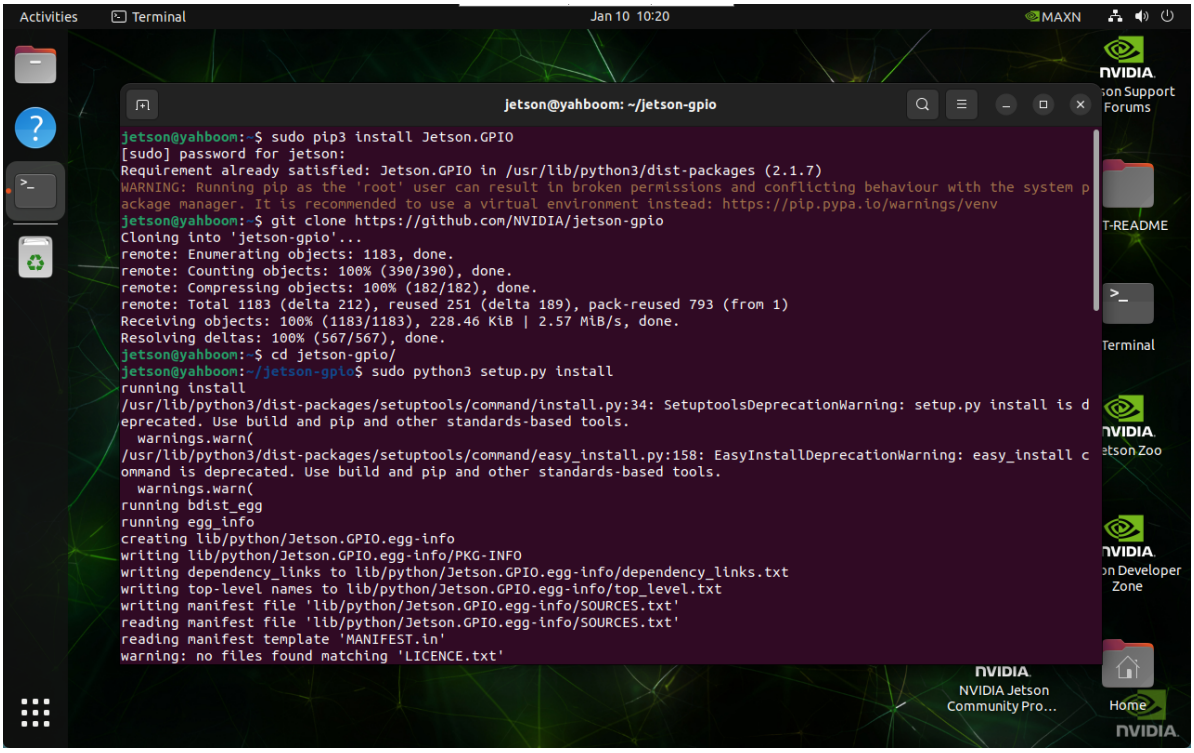
1.2 Manual Installation

It is recommended to use the automatic installation method. Manual installation may result in an outdated version.

```
git clone https://github.com/NVIDIA/jetson-gpio
```

```
cd ~/jetson-gpio/
```

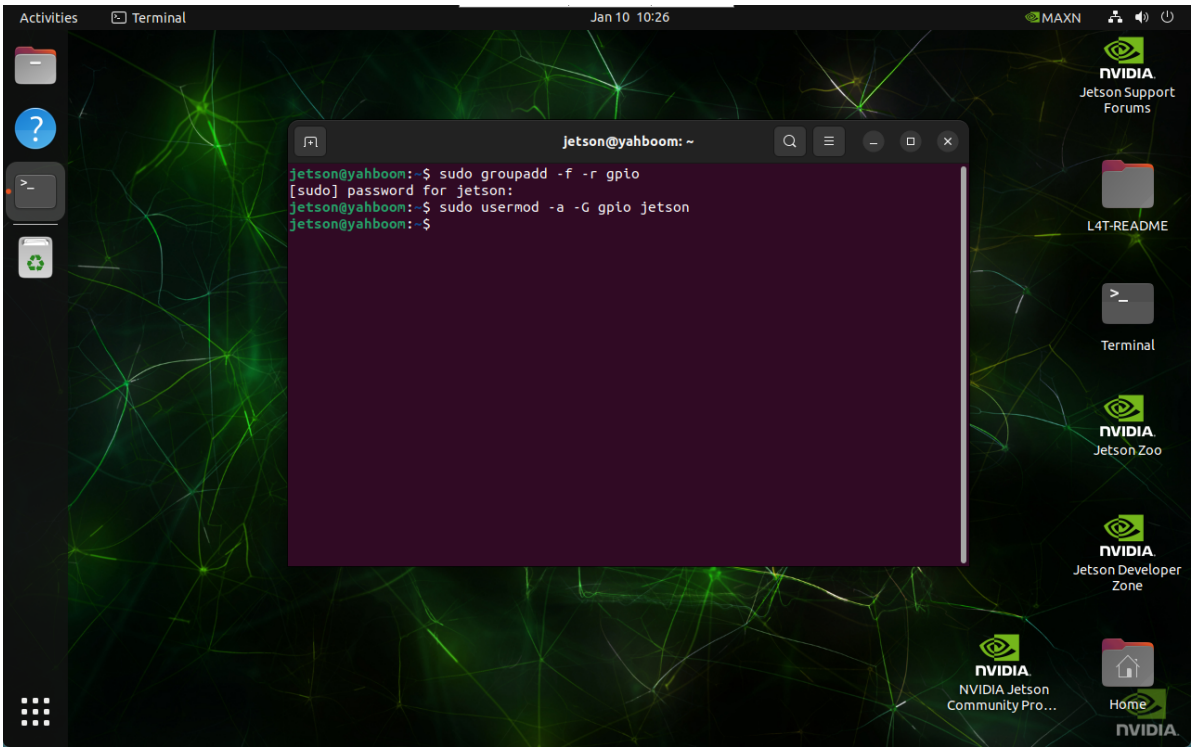
```
sudo python3 setup.py install
```



2. Set User Permissions

Allow the current system user to access and use the Jetson.GPIO library: where `jetson` is the system username

```
sudo groupadd -f -r gpio
sudo usermod -a -G gpio jetson
```



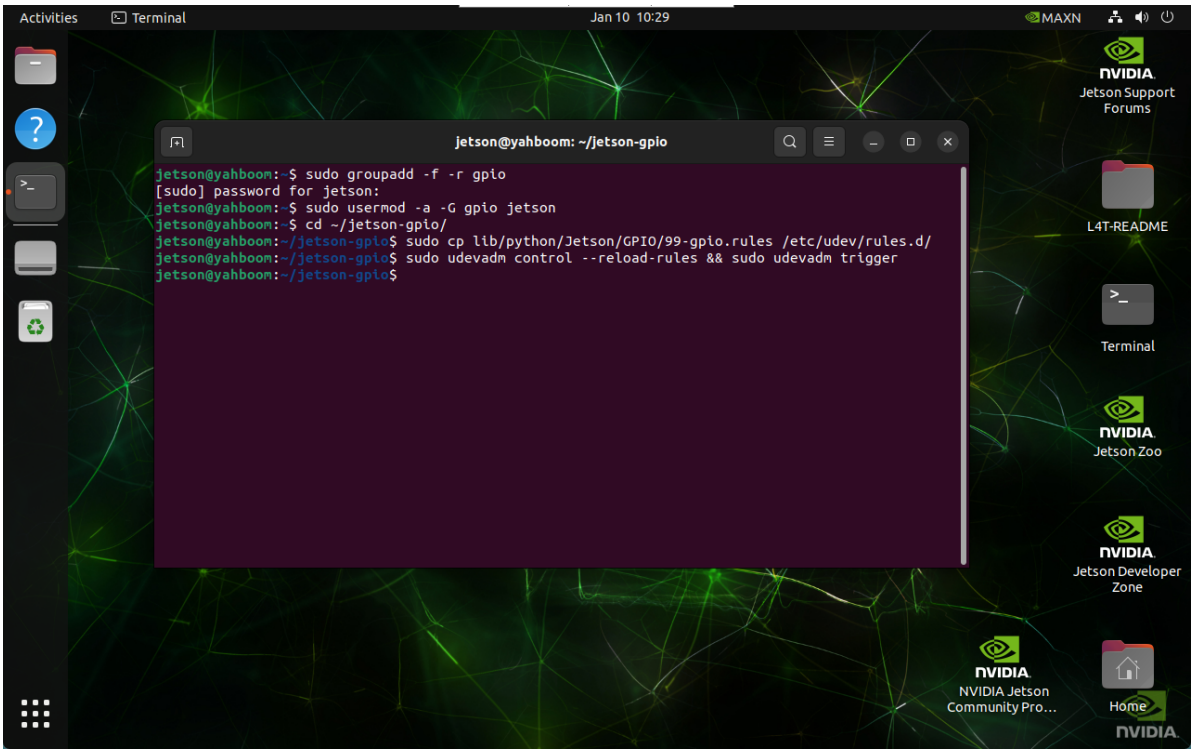
3. Customize the Rule File

3.1 Copying the Rule File

```
cd ~/jetson-gpio/
sudo cp lib/python/Jetson/GPIO/99-gpio.rules /etc/udev/rules.d/
```

3.2 Reloading udev Rules

```
sudo udevadm control --reload-rules && sudo udevadm trigger
```



4. Setting the Motherboard Model

Currently, Jetpack 6.1 does not pre-set the motherboard model. Before each GPIO control operation, you need to set the motherboard model in the terminal:

```
export JETSON_MODEL_NAME=JETSON_AGX_ORIN
```

5. References

https://github.com/NVIDIA/jetson-gpio